

Bluff

YOUR WAY THROUGH

NLP

Natural Language Processing (NLP) is an Artificial Intelligence (AI) undertaking. It aims to process sentences written in a language such as English so as to extract meaning from them and represent this meaning in a manner that a computer can understand. If you want to portray yourself as an AI practitioner, simply say that amongst all AI disciplines, you like NLP best and then go on to describe how you do it.

How do you make a computer understand English?

To this, you reply, "That's what NLP is for. You break up the sentences into parts of speech, and then parse it just as you would any other string, of course, ellipsis is one major hurdle." ('Ellipsis' means an instance where the context is not available.)

What is ellipsis? Intimidate your audience here. "You don't know what ellipsis is? Say you reply to a question about your name with 'John Smith'. Here, you haven't specified what John Smith is and that's ellipsis."

Your audience will probably ask you at this point, "So that's all there is to NLP? Parts of speech and parsing?"

Since you don't know too much yourself, you launch off into unfamiliar words— "No, no, there's actually much more. You have got to consider things like failure of substitutivity, you have to learn to use the FOPC and HMMs, construct working grammars and much more."

What is FOPC and HMM?

Remember to learn the full form—First Order Predicate Calculus. It sounds nice and AI-ish. Of course, you deliberately use the abbreviations in the first instance, so you can expand them later. HMM stands for 'Hidden Markov Model'.

Finish off with: "Like I said, it's basically the manipulation of the parse tree. You construct the tree and do all sorts of things with it. I can direct you to a tutorial if you like."

Nuptials Time!

Two distinguished gentlemen of the *Digit* team are no longer single; they have been sacrificed at the altar of marriage. It is a sacrifice they have invited upon themselves, having fallen head over heels in love with two lovely young women.

Copy Editor Robert Sovereign-Smith aka Raaabo married Karen at a ceremony in which no one could recognise Robert. He had worn formal clothes for the first time in his life since he'd peddled goods as a salesman for a leading company. Heck, he'd even shaved for the wedding reception.

DIGIT DIARY

Jayan, *Digit's* lead designer, who had kept his beau Shiny hidden from everyone's knowledge by cleverly wooing her when he was in Kerala, did no such thing. His facial hair, the pride of his Malayalee clan, had been groomed with as much discipline as he puts Photoshop streaks in his opening visuals. His beard represents his personality, as it did for Osho, G B Shaw and other hirsute champions of history.

Both Robert and Jayan are back at work now, with all the vim and vigour of newlyweds. We are now waiting for the seven-year itch!

XML SMELL

A Language For Smell

A researcher at Huelva University in Spain claims to have created a version of XML—often called the successor to HTML—that can transmit fragrances. The ability to transmit smells could have applications in several fields including tourism, gaming, marketing, medicine, and so on.

The university has worked with researchers in both computing and chemistry to come up with the concept of the XML Smell language. It said the initial idea was to print smells on paper using laser and inkjet printers, but in this approach, the fragrances degraded over a short period of time. The scientists now claim to have developed a dual-pass printing system, using two polymers which degrade by light and by touch, so that the smell only emerges when a document is read.

Not content with this, researchers quickly realised that smells could be propagated over networks and, of course, the Web. They therefore created XML Smell, which they claim can define in a universal and standardised way the

transmission of smell. This will allow for the transmission of fragrances by e-mail, by SMS to a mobile phone, or via a TV show.

Currently, the scientists are designing a device which will sit close to a TV, a radio, a phone or a PC, and which contains a 'smell palette'. The components in the palette get manifested according to instructions in the XML Smell language.

ROBO-CUP

Robots Will Win World Cup By 2050!

A Japanese consortium of robotics experts has

Ishiguro and his team are talking about the successors to a robot called *VisiON*, which is just 38 cm tall and weighs only 2.4 kg. *VisiON* operates independently of human input, makes its own decisions based on information it perceives and recognises the football, approaches it and delivers a hefty kick.

It can identify an opponent and shield the ball. Atop its head is a 360-degree vision sensor, so it does not have to turn its head at all.

Robot World Cup tournaments have been with us for quite some time now—the most recent being held in Lisbon. RoboCup 2005 is scheduled for July in Osaka, Japan.

Ishiguro is confident that players of steel and wire will one day lift the



claimed that their engineered humans will outplay real ones—in football—within 45 years.

World Cup. "The important thing to remember is what you see here is just the beginning," he says.